

Optimal Treatment Methods for Breast Cancer Patients and Involvement of the Estrogen Receptor Gene (ESR1) in Method Choice

Abstract: Breast cancer is the most common malignancy for women worldwide. This study uses data obtained by the Netherlands Cancer Institute to investigate the optimal treatment methods for breast cancer patients with varying expression levels of the estrogen receptor (ESR1) gene. The goal is to identify these methods and to see if a difference exists in the choice of optimal treatment combination for patients with lower or higher expression levels of the ESR1 gene. Statistical testing confirmed a significant difference in indicated survival time (IST) for our two groups, and a polynomial regression model detailed the nature of this difference. Further information was derived from a Pearson Correlation Matrix, which developed understandings on the dataset overall. Mean IST was plotted for our subsets for each therapy per cancer grade to observe differences in methods. It was discovered that high ESR1 expression is a positive prognostic indicator in general, but more so for late-grade cancer. For grade 1 cancer, we found no difference in the optimal method between ESR1 groups (chemotherapy and amputation together) or for grade 2 (chemotherapy). However, a difference was observed for grade 3 (chemotherapy and amputation together for high ESR1, hormonal therapy for low ESR1), indicating that ESR1 expression level is significant in the choice of treatment combinations for a patient, particularly for a later grade cancer.

Introduction: Breast cancer is a form of cancer which develops specifically in the cells of breast tissue. It has been estimated to be curable in ~70-80% of cases with early-stage identification, but advanced breast cancer is deemed incurable with current available therapies (Harbeck et al., 2019). Breast cancer's development has long been known to be associated with the steroid hormone estrogen – it is one of the primary hormones which influences breast cancer's prognosis (Sommer et al., 2001). A high expression level of this gene can be a positive prognostic parameter in breast cancer, particularly in the later stages - in early breast cancer, no receptor-associated biomarker exists that identifies patients with a particularly more favourable outcome (Singer et al., 2022).

We will look at data on 272 unique breast cancer patients obtained by the Netherlands Cancer Institute (NKI), a leading establishment in Amsterdam. Data was procured from Data World (Data World: NKI Breast Cancer Data). The exact demographics relating to patients in our data are unclear – it is not explicitly stated if all data was collected from the same hospital/ country. However, it was likely all collected at the NKI. It is also not indicated if this data pertains to both male and female cases of breast cancer. However, most studies and datasets focus on women specifically, so the patients in this dataset are likely women.

This report aims to analyse trends in breast cancer outcomes in relation to the expression level of the ESR1 gene. Specifically, we will look at determining optimal methods of treatment and if there is a difference in which methods are best suited for patients with lower or higher expression levels of ESR1. The methods of treatment we will be focusing on are chemotherapy, hormonal therapy and amputation. This research topic has been chosen because analysing trends for survival time within our dataset may prove beneficial in identifying optimal treatment methods for specific groups of patients. Identifying these methods could have real-life impacts and assist with therapy choices for patients, potentially helping to increase survival time.

To assess the relationship between the expression level of the ESR1 gene and indicated survival time (IST), we must think deeper into the process of choosing treatment methods for a patient. Initially, one may conclude that receiving more treatments should positively impact survival outcomes; however, we

must consider the fact that these treatments are not by themselves ‘healthy’ (Macmillan Cancer Support: Late effects of cancer treatment), so doctors will be hesitant to utilise every method on a patient to avoid negatively impacting survival outcomes. Treatment methods are chosen through complicated analysis of a patient’s circumstances (Cancer Research UK: Stages of cancer) – methods are highly individualised, and so there may be no universally optimal approach. We can look at how combinations of different therapies generally affect the survival outlook for all patients. However, we should be looking at how severe the cancer is in determining which method to use - the severity and staging are essential in selecting a suitable treatment method (Cancer Research UK: Stages of cancer). Our data has multiple columns that relate to this type of information: the grade of cancer, tumour size, number of nodes, level of lymphocytic infiltration, and the level of blood vessel/ lymph cell infiltration. Since we want to look at how the overall severity of cancer affects our optimal choice of combination of treatment methods, we will focus on the grade of cancer. The cancer grade is determined considering all the factors of the other preceding columns – it is the most encompassing category when it comes to classifying overall severity by definition (Cancer Research UK: Stages of cancer). With this, we can define our final research question: “What are the optimal treatment methods for different severities of breast cancer, and does the expression of the ESR1 gene make a difference in choosing such a method?”.

Methods:

Explorations and Alterations of the Dataset

The original dataset contains information about the patients, the severity of their cancer, and details about the expression levels of over a thousand genes. Some of these column’s titles could be more intuitive or are unnecessary in the pursuit of answering our research question. For this reason, the dataset has been altered to avoid confusion and increase clarity in figures.

Data was imported into Python, and issues were thoroughly checked for – no missing data, misplaced data, value problems, or duplicate records were found. In addition, a manual inspection of the dataset was also undertaken, and the same results were obtained. Data wrangling was performed to tidy the dataset:

- Unnecessary columns were removed for easier traversal through the dataset.
- Data not under inspection was dropped.
- Column titles were altered to portray them understandably and improve interpretability.

Our primary focus is on IST, the expression level of the ESR1 gene and the patient's cancer grade, so these columns were kept. However, we have retained other columns to help shape initial understandings of our dataset and to see if sensible correlations are observed. Two columns were added for extra analysis: "Different Therapies" and "Treatment Combinations". Information on these columns and their relevance is detailed in Table 1, alongside the rest of the dataset (and alterations).

Table 1: Raw Dataset to Working Dataset Details (*CN* = “*Changed Name*”, *MDL* = “*More descriptive labelling*”), *MI* = “*Do not need multiple identifiers*”

<i>Column Names (Old)</i>	<i>Changes</i>	<i>Type</i>	<i>Explanation</i>
Patient	Dropped	String	MI
ID	N/ A	Integer	Defines patient ID. Kept as it is useful to have an identifier for each patient.
age	CN: “Age”	Integer	The ages of patients in years.
eventdeath	CN: “Death	Integer (0 or 1)	Indicates survival status. Containing values 0 or 1 (0 = false and 1 = true; note this is the same for columns with same format).

survival	CN: “Indicated Survival Time”	Float	MDL. Details the presumed survival time for a patient in years.
timerecurrence	CN: “Indicated Recurrence Time”	Float	MDL. Details the presumed cancer recurrence time (how long it is estimated until the cancer returns) in years.
chemo	CN: “Chemotherapy Used”	Integer (0 or 1)	MDL. Whether chemotherapy has been used on the patient.
hormonal	CN: “Hormonal Therapy Used”	Integer (0 or 1)	MDL. Details if hormonal therapy has been received.
amputation	CN: “Amputation Used”	Integer (0 or 1)	MDL. Indicates if breast amputation has been performed.
histtype	CN: “Histological Type”	Integer (1 to 7)	MDL. Indicates the histological type of a patient, which describes the type of tissue that the cancer has originated in (SEER: Cancer Classification).
diam	CN: “Tumour Diameter”	Integer	MDL. Indicates the main tumour’s diameter in millimetres.
posnodes	CN: “No. of Nodes”	Integer	MDL. Indicates the number of nodes (tumours) present in a patient.
grade	CN: “Cancer Grade”	Integer (1 to 3)	MDL. Indicates the grade of cancer a patient has. Grades range from 1 to 3 and are an indicator of the severity of cancer a patient has, with 3 being the most severe.
agioinv	CN: “Blood Vessel/ Lymph Cell Infiltration Level”	Integer (1 to 3)	MDL. Indicates the extent of invasion of cancer into blood cells.
lymphinfil	CN: “Lymphocytic Infiltration Level”	Integer (1 to 3)	MDL. Indicates the extent of invasion of cancer into lymphocytes.
barcode	Dropped	Integer	MI
esr1	CN: “Expression Level of Estrogen Receptor Gene”	Float	MDL. This column contains a float value describing the expression level of a patient’s ESR1 gene. Lower values indicate less expression from the gene, whereas higher values indicate a higher expression level.
Other Gene Expression Levels	Dropped all	Floats	They have been dropped because interpreting results from all of these genes is beyond the scope of this paper.
N/ A	Created column “Different Therapies”	Integer (0 to 3)	Details the total number of different therapies used on a patient. Every value in the column is initialised at 0. Then, for each row, we check the values for each therapy (0 or 1) and add them to get the values for this column. The purpose of this is to assist us with developing an understanding of how the number of therapies received impacts different factors within our dataset (e.g. the correlation with IST).

N/ A	Created column “Treatment Combinations”	String	Similar to “Different Therapies” except we are creating a string detailing which therapies were received by adding together defined strings based on the values in each therapy column (e.g. chemotherapy = 1, hormonal =1, amputation = 0 gives us “Chemotherapy & Hormonal”). This column was introduced to help us subset our data to gain insight on IST trends for specific combinations of therapies.
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Upon completing alterations to the dataset, distributions of relevant categories were examined to gain an overview of the shape and structure of our dataset. Histogram plots were created to enhance understanding of this (Figure 1a and 1b). These plots indicate that our data is not normally distributed for both IST and for the expression level of the ESR1 gene, suggesting for statistical analysis, non-parametric methods may be preferred. We also do not observe any unusual spikes or dips.

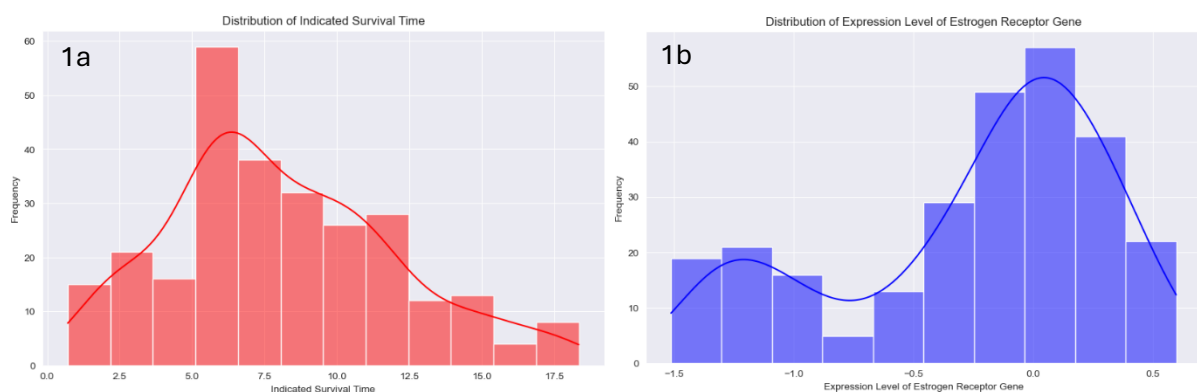


Figure 1a & 1b: Distributions of Indicated Survival Time (1a) and Expression Level of Estrogen Receptor Gene (1b), depicted with KDE (kernel density estimation) line

Statistical Testing

We are assessing whether there is a significant difference in IST for those with a high expression level of the ESR1 gene compared to those with a low expression. To achieve this, data was split into two subsets: “Low ESR1 Expression Group” and “High ESR1 Expression Group”. The median of the ESR1 expression level was calculated, and patients with expression levels lower than this were placed in the first category. In contrast, patients with higher expression levels than the median were placed in the second category.

We need to learn more about our actual data distributions to perform a relevant statistical test. Gaining this information will help solidify a choice of an appropriate parametric/ non-parametric test to check for significant differences between our two ESR1 groups.

It was checked to see if a parametric or non-parametric method was better suited to our data by analysing the normality of the two sets. Based on our distribution for ESR1 in Figure 1b, we may presume that our split groups would also suit a non-parametric test, but to ensure this, another visual inspection was undertaken alongside the utilisation of a Shapiro-Wilk test. This hypothesis test evaluates whether data in a set is normally distributed. Our null hypothesis (H_0) is that the data set is normally distributed, and a low p-value (<0.05) rejects this H_0 in favour of the alternative hypothesis (H_1), that our data is not normally distributed. The results from this test will help us determine whether our data is better suited to an Independent Two-Sample T-Test (parametric) or a Mann U Whitney Test (non-parametric). Then, the results of these tests will tell us if there is a significant difference between the IST for our two ESR1 groups. Here, a low p-value rejects H_0 : “There is no significant difference in IST for our ESR1 groups”, and a high p-value accepts H_0 , disregarding H_1 : “There is a significant difference in IST for our ESR1 groups”.

Polynomial Regression

Alongside our statistical testing, we can create a regression model between ESR1 expression and IST. Observing that there is a significant difference between the two sets does not give us information about this difference – a model will allow us to obtain a graphical interpretation of the relation between the two aspects as ESR1 levels vary. We assume that as the expression level of ESR1 increases, we see an increase in IST, so it could make sense to use a linear regression model. However, results from a preliminary scatterplot (Figure 2) ruled this out.

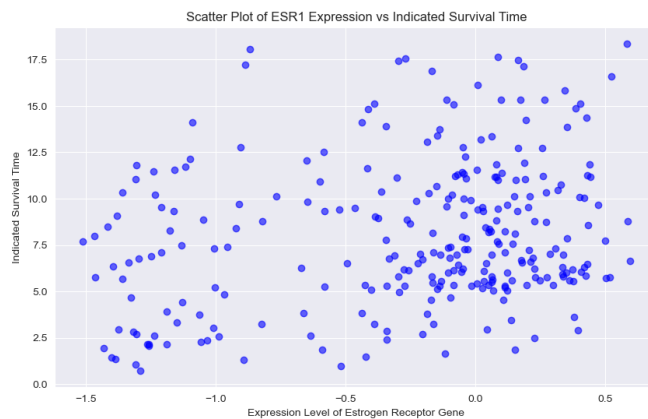


Figure 2: Scatterplot of ESR1 Expression vs Indicated Survival Time

Although survival time does seem to increase with a higher expression of ESR1, the relationship seems harder to pinpoint due to the diversity in the scatterplot—it does not seem to be just linear. To capture the nuances of our data, we will use a polynomial regression model instead. We will fit our model using degree 5 (this number was chosen to avoid underfitting/ overfitting after testing with lower/ higher degrees).

Pearson Correlation Matrix

Initial explorations of the dataset also involved looking at the correlations between all different aspects of the data. The Pearson Correlation Matrix can summarise a large data set and identify patterns, which, when observed, can help us decide which direction to take research. An annotated heatmap of our dataset was generated in Python using the Pearson Correlation Coefficient (PCC), which assesses the strength of the relationship between two aspects of a dataset to get a value indicating correlation. A positive PCC indicates a positive relationship, whereas a negative value indicates a negative relationship. For example, if we have a 0.5 PCC for cancer grade and death, then that means as cancer grade increases, we will see death increase, but a -0.5 PCC for cancer grade and survival time would mean that as cancer grade increases, we see survival time decrease. The generation of this matrix will be crucial in developing our initial understanding of the data set. It will additionally advance our understanding of how all aspects of the dataset interact with each other. We have not included the column "ID" in this analysis as it provides no information. "Treatment Combinations" has also not been included here due to the nature of the values in the column, as we cannot calculate relationships with a string.

Parsing and Plotting by Grade for Analysis

Once we have established all of the above, we can move on to answering our final research question. For the following groups: Whole Dataset, Low ESR1 Individuals and High ESR1 Individuals, we will split each by the possible cancer grades for nine sets (3 grades for each group). We will then create a bar plot utilising the column "Treatment Combinations" we created to obtain a graphical interpretation of the mean IST for individuals who received different types of therapies within specific grades for a particular group. This information will be vital in helping us discern the best treatment method for individuals who fall into our ESR1 groups with different severities of cancer, and we can compare if there is a different result for each group.

Results:

Statistical Testing

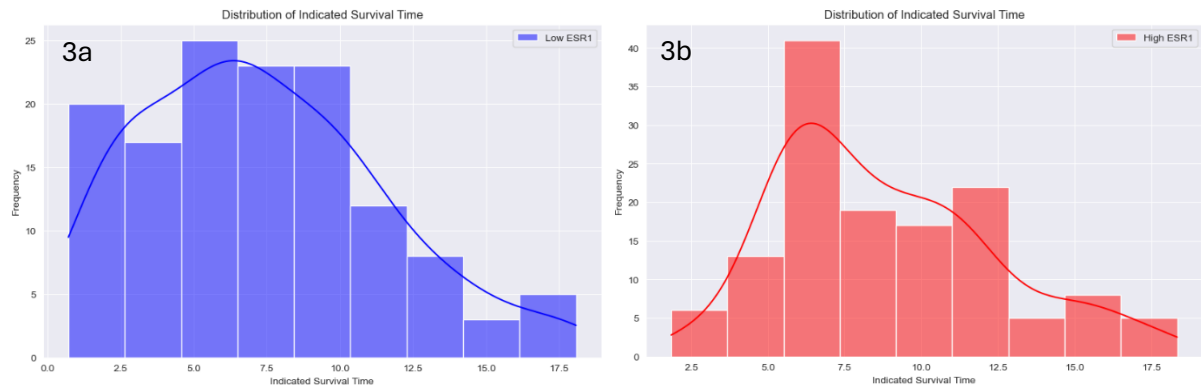


Figure 3a & 3b: Distributions of Indicated Survival Time for Low ESR1 (3a) and High ESR1 (3b), depicted as histograms with general line

We gain the initial impression that our distributions are both non-normal, suggesting that a non-parametric method would be best suited. Upon performing our Shapiro-Wilk test, we confirm this, seeing our p-values as < 0.05 from Table 2. We reject H_0 in favour of H_1 for both groups: our data is not normally distributed in either set.

Table 2: Shapiro-Wilk Test Results for ESR1 Groups Indicated Survival Time

Indicated Survival Time	P-values for Shapiro-Wilk Test
Low ESR1 Expression Group	0.00114118
High ESR1 Expression Group	0.00007027

As we have determined that our data is non-normally distributed, we will proceed to use the Mann-Whitney U Test to determine if our IST is significantly different between the two ESR1 groups. Our Mann-Whitney U Test returns a p-value of 0.00112745, which indicates we should reject H_0 and that there is indeed a significant difference between our two groups for the IST.

Polynomial Regression

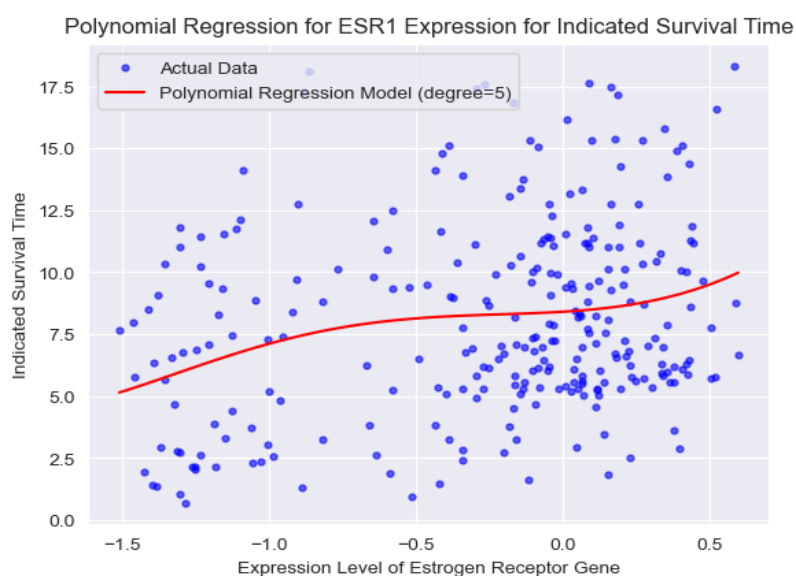


Figure 4: Polynomial Regression of ESR1 Expression for Indicated Survival Time

Our polynomial regression indicates that as the expression level of ESR1 increases, the IST will increase. Notably, we observe a non-linear relation – a steady increase, levelling off in the middle third with a slight plateau effect. The endpoints for our curve are situated at significantly different points (5.15, 9.98), highlighting the significance of ESR1

expression in better IST outcomes. Our model seems to fit well using degree 5 - we do not see any

spikes or an unexpected slope, and what we observe aligns well with the initial ideas developed from our research.

Pearson Correlation Matrix

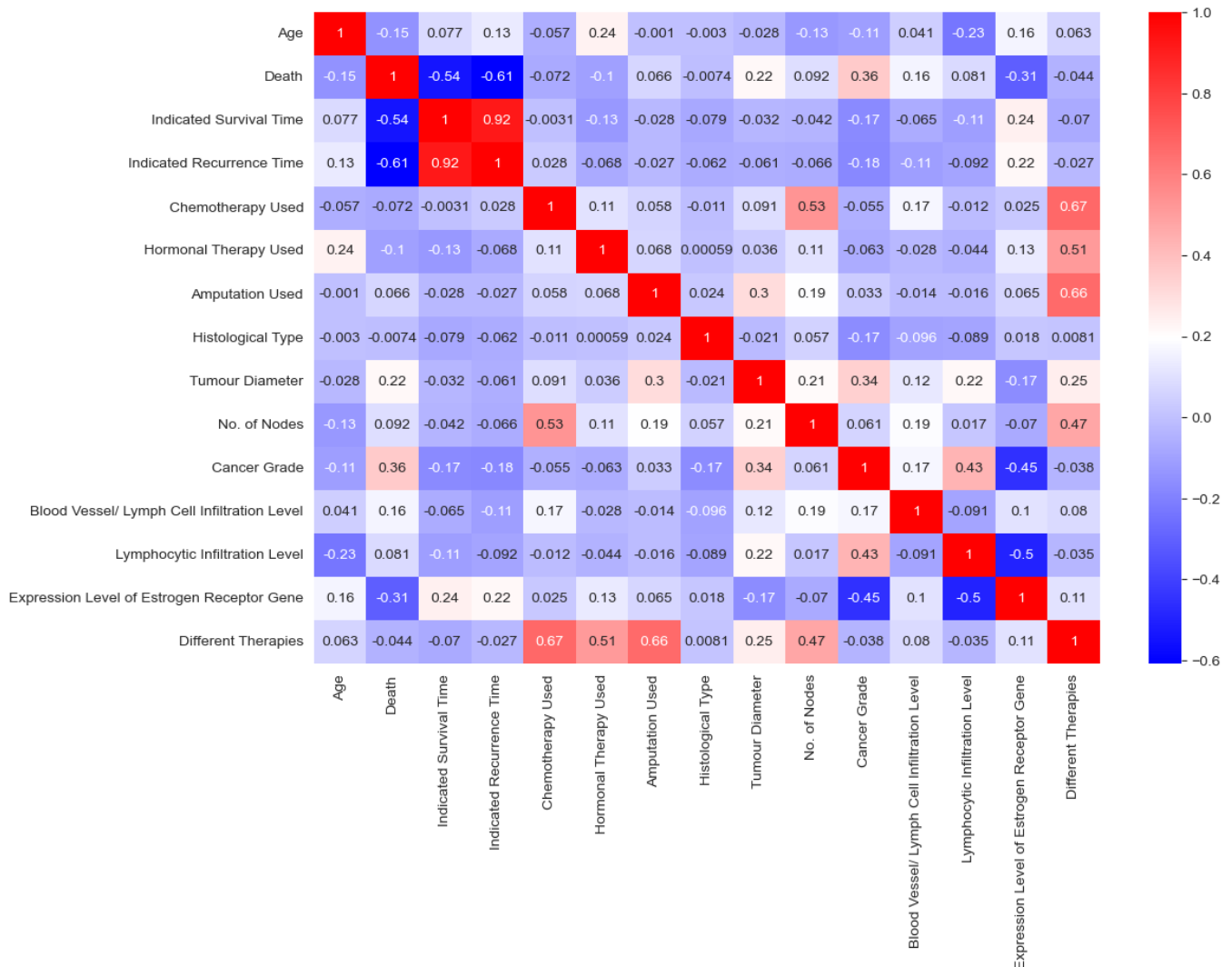


Figure 5: Pearson Correlation Matrix (Working Dataset excl. “ID” and “Treatment Combinations”), Colour bar on RHS indicates the strength of correlation – 0 indicates no correlation, moving up (redder) or down (bluer) indicates a stronger positive or negative correlation respectively

Significant correlations were found in multiple areas of interest: there is a strong negative PCC (-0.5) between the expression level of the estrogen receptor gene and lymphocytic infiltration level, as well as cancer grade (-0.45). There is also a moderately strong negative PCC (-0.31) with death and a positive PCC (0.24) with IST. These results indicate that having a high expression of the ESR1 gene can correlate with a higher IST and a better outlook for multiple traits involved in breast cancer. These findings lead us to the understanding that having a higher expression level of the ESR1 gene leads not only to higher expected survival outcomes but also to a better outlook for multiple of the different aspects which play a part in the severity of cancer.

We can see that cancer grade correlates strongly with death (0.36), which means it is definitely a sensible choice to classify severity by. We also see a high PCC with tumour diameter (0.34) and lymphocytic infiltration level (0.43), suggesting that these two are important in determining a patient’s cancer grade.

Inspecting the correlation outcomes from our added column “Different Therapies”, we can see no correlation with a better survival time. This aligns with what we discussed earlier on – more treatments

likely mean a more problematic cancer. We see a positive correlation (0.25) with tumour diameter, as well as with the number of nodes (0.47), implying that the reason we are not observing a positive correlation with survival time (-0.07) here is that the cancer is indeed worse in these scenarios, hence why multiple treatments have been utilised in the first place.

Parsing and Plotting by Grade for Analysis

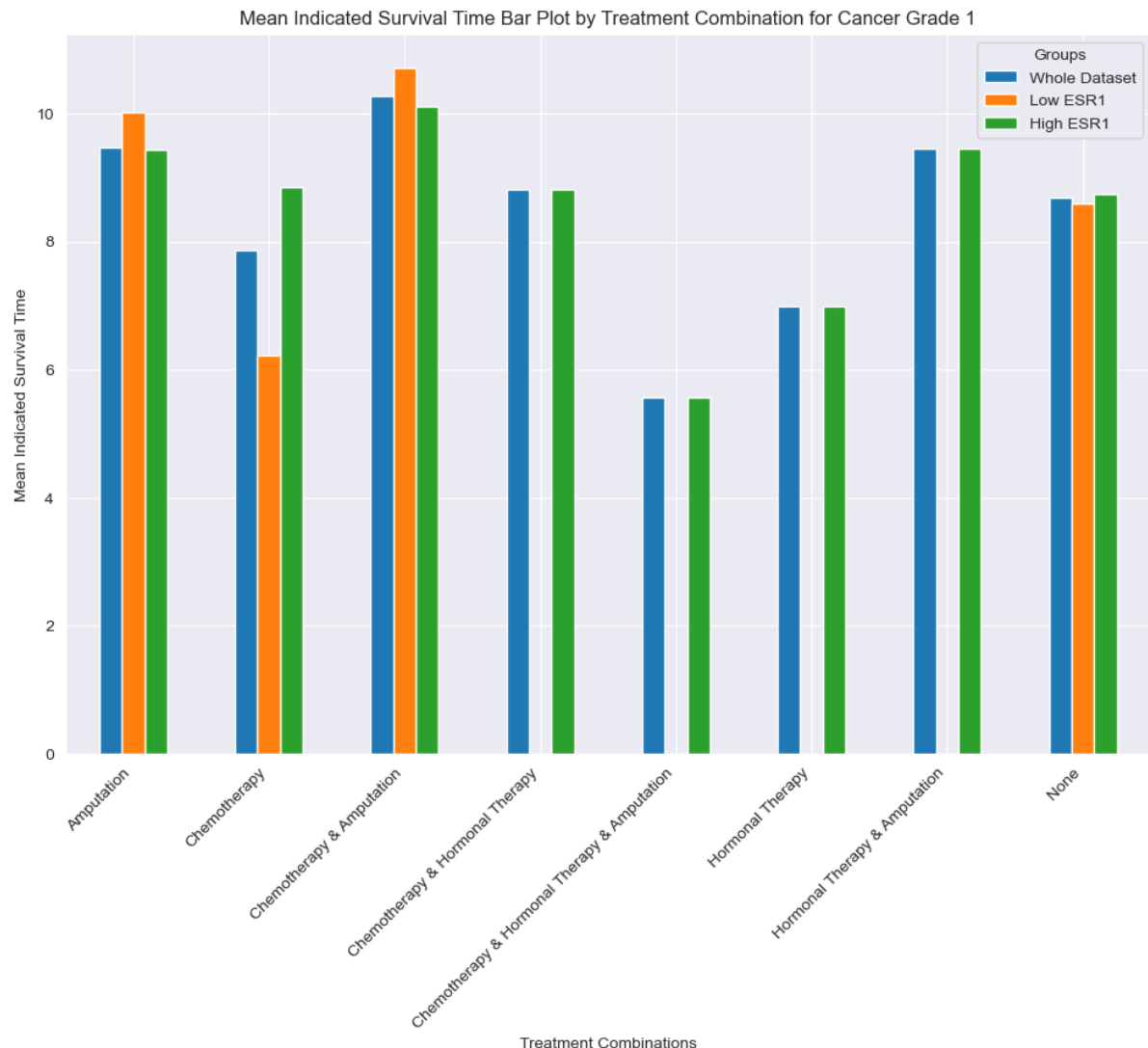


Figure 6: Bar Plot of Mean Indicated Survival Time by Therapy Combination for Grade 1 Cancer

We observe some variance here in the set with the highest mean IST – for some treatment combinations, we see lower ESR1 having a slightly higher mean. Our results here make sense based on our research – more favourable outcomes are generally discovered in later-stage cancers. However, we do observe some significant differences in the mean ISTs for chemotherapy by itself, with high ESR1 definitely being beneficial here. We observe that all subsets have the same most effective therapy combination,

chemotherapy and amputation used together. This tells us that the observed optimal treatment method may not differ for grade 1 cancer for our groups.

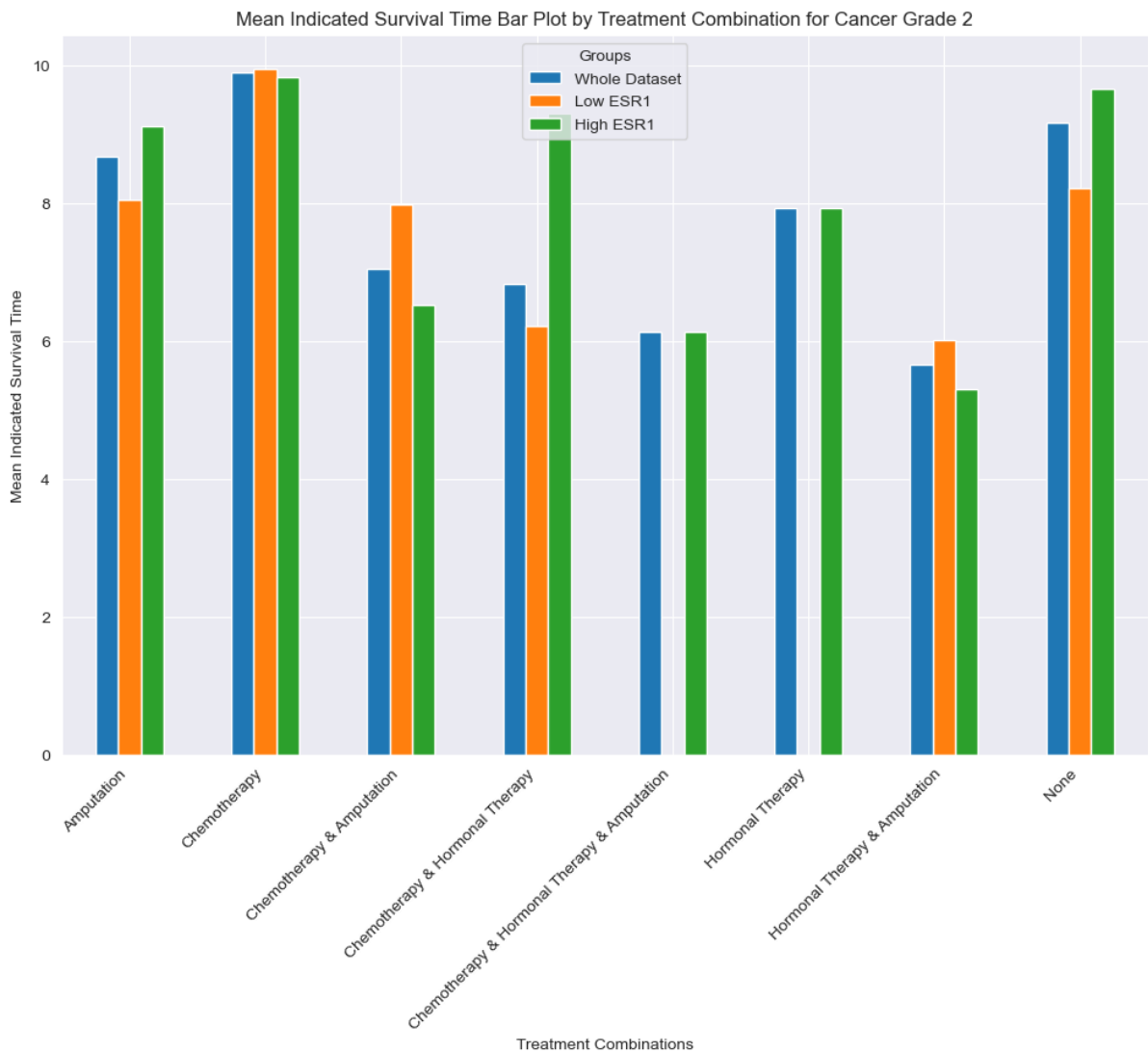


Figure 7: Bar Plot of Mean Indicated Survival Time by Therapy Combination for Grade 2 Cancer

This time, we observe chemotherapy as the optimal treatment for the IST of all groups. We also observe in some treatments that lower ESR1 has a higher IST, but again, minimally. This still does not conflict with our earlier research, as we expect the most prominent results for grade 3. For the majority of treatments, high ESR1 is preferable. We can see that, in general, the mean IST has decreased; this makes sense since we are looking at a more severe cancer this time. It is also worth noting that there is a significant difference in the "None" category for the two groups, highlighting how higher ESR1 by itself can be beneficial in survival outcomes.

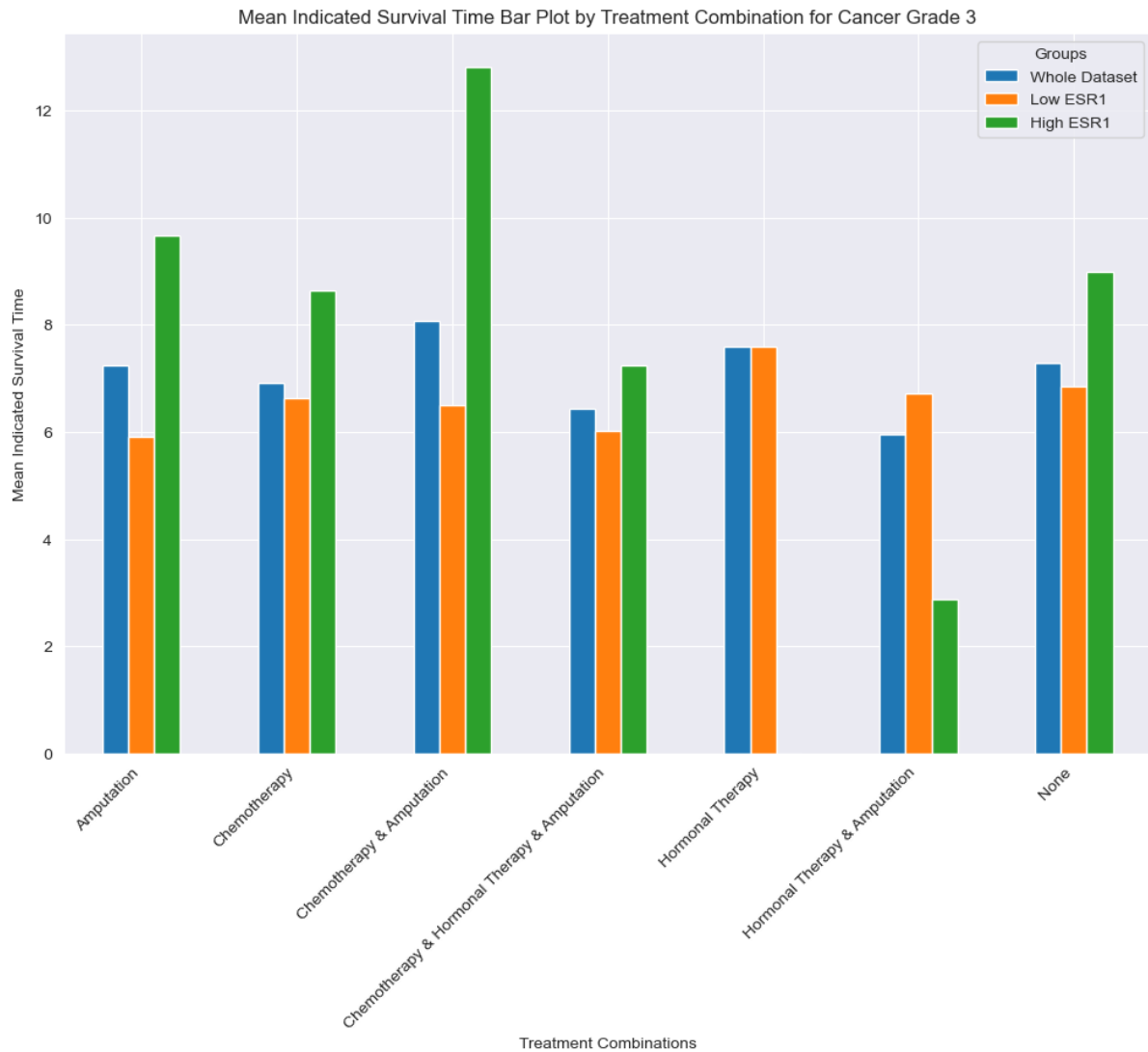


Figure 8: Bar Plot of Mean Indicated Survival Time by Therapy Combination for Grade 3 Cancer

Here, we observe the most significant differences in our two groups and observe differences in the optimal treatment methods. High ESR1 patients have a much higher IST for every therapy (where there is data for the group) except for hormonal therapy and amputation used together. We see chemotherapy and amputation together as the best treatment combination for high ESR1 patients (and for the whole dataset overall). However, hormonal therapy is the optimal method for those with lower ESR1 expression levels. This heavily aligns with current understandings of ESR1's effect on survival outlooks in late-stage cancer.

Discussion

We have observed a correlation between IST and high ESR1 expression, alongside a decreased correlation with multiple negative factors for cancer (e.g. grade, number of nodes). Our statistical analysis showed a significant difference in IST for low ESR1 patients and high ESR1 patients. This idea was then developed, and our polynomial regression model provided a graphical interpretation of this relationship. It aligned with our results from our Pearson correlation matrix, and we observed that as ESR1 expression increased, so too did the IST in general. The combined results of these methods have allowed us to gain insight into the relationship between ESR1 and IST. A higher expression level is likely linked with better survival outlooks.

Finally, when plotting IST by therapy combination for each cancer grade, it is observed that high ESR1 increases the IST again. We observe the starkest differences for grade 3 cancer, implying that although high ESR1 in general seems beneficial for IST, it has the most impact on further progressed cancers. We observe, overall, that the combination of chemotherapy and amputation is the most frequently occurring treatment method with the highest IST. However, this varies for grade 2 cancers, where we found that chemotherapy alone was the optimal method, and in grade 3, where we see that lower ESR1 patients had the highest survival rates for hormonal therapy alone instead.

Conclusion

We have utilised our dataset to provide insights on the optimal treatment methods for different severities of breast cancer and observed how the expression level of the estrogen receptor gene affects such optimal method and the indicated survival time as a whole. Correlation matrices, statistical analysis, polynomial regression, and parsing were used to obtain these results. The information found aligned with current understandings of how the ESR1 gene influences IST - we found that there were indeed some differences in the optimal treatment methods within different cancer grades for our two groups of ESR1 expression, specifically in further progressed cancers. We saw some overlaps of optimal methods in lower grades. In grade 1 cancer, chemotherapy and amputation together were observed to be the optimal combination across the board. Whereas, in grade 2, it was chemotherapy. However, for grade 3, we observed that chemotherapy and amputation together were best for individuals with high ESR1 expression, whereas the lower ESR1 expression group saw improved IST from hormonal therapy. We found that, in general, high ESR1 can be beneficial for IST, but particularly more so in these developed cancers, and that there was a significant difference in treatment choice for our two groups within this grade.

References

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Appendix A:

Annotated Python Code (MT4561_Project_Code.ipynb), included in zip file